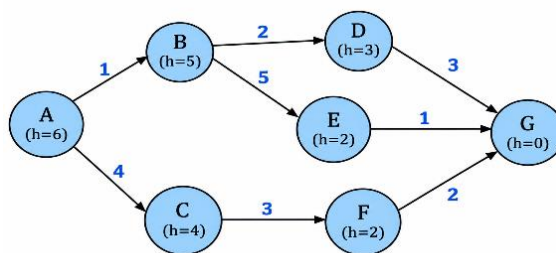


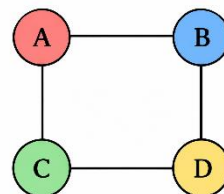
Q1) Write a program to implement Breadth First Search (BFS) for the Monkey Banana Problem. **[10 marks]**

Q2) Write a program to implement the A* Search Algorithm to find the shortest path between a source node and a destination node **[20 marks]**



OR

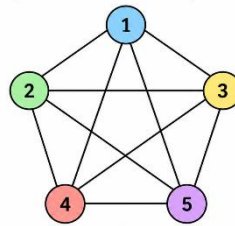
Q2) Write a program to solve the Map Coloring Problem using the Constraint Satisfaction Problem (CSP) approach. **[20 marks]**



Q3) Viva **[05 marks]**

Q1) Write a program to solve the Water Jug Problem using BFS. **[10 marks]**

Q2) Write a program to implement the Backtracking Algorithm for solving a constraint satisfaction problem. **[20 marks]**



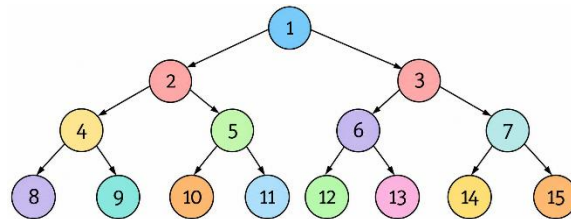
OR

Q2) Write a program to implement a Game Tree Representation for a two-player game. **[20 marks]**

Q3) Viva **[05 marks]**

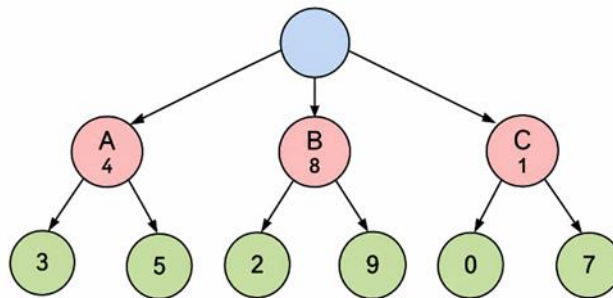
Q1) Write a program to implement BFS traversal for a graph.

[10 marks]



Q2) Write a program to implement Alpha-Beta Pruning for optimizing the Minimax search process.

[20 marks]



OR

Q2) Write a program to implement Propositional Logic and evaluate logical expressions using operators such as AND, OR, and NOT.

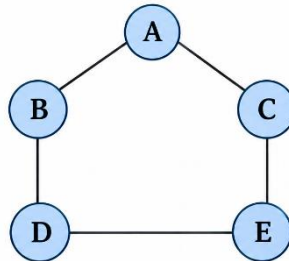
[20 marks]

Q3) Viva

[05 marks]

- Q1)** Write a program to solve a state-space search problem using BFS. **[10 marks]**

Start Vertex = A



- Q2)** Write a program to implement Predicate Logic using predicates, variables, and quantifier. **[20 marks]**

OR

- Q2)** Write a program to implement Gaussian NB classifier with 3 classes on numeric data, use `make_blobs` to create a dataset X of 200 sample points in 2D Space, 3 clusters and each cluster spread (SD) over 1.5. For testing (Prediction), use the following values to test the Gaussian NB Classifier. **[20 marks]**

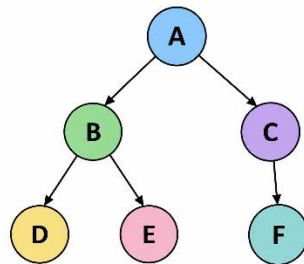
Input (Values) : [-2, 5], [0,0], [6, -0.3]

Output : [0 1 1]

Print the accuracy score using the `accuracy_score()` from `sklearn.metrics` library..

- Q3)** Viva **[05 marks]**

- Q1)** Write a program to implement Depth Limited Search (DLS). **[10 marks]**



- Q2)** Write a program to implement Bernoulli NB classifier with 2 classes on numeric data, use `make_classification` to create a dataset X of 300 sample points in 2D space with the following parameters :

`n_features=2, n_informative=2, n_redundant=0, n_classes=2.`

For testing (Prediction), use the following values to test the Bernoulli NB Classifier.

Input (Values) : `[[0, 0], [0, 1], [1, 0], [1, 1]]`

Output : `[0, 1, 0, 1]`

OR

- Q2)** Write a program to implement a Support Vector Machine (SVM) classifier on the Iris dataset. **[20 marks]**

Input (Dataset and Class Labels):

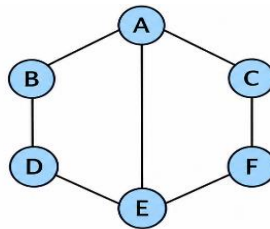
`X = [ [5.1, 3.5, 1.4, 0.2], [4.9, 3.0, 1.4, 0.2], [4.7, 3.2, 1.3, 0.2], [4.6, 3.1, 1.5, 0.2], [5.0, 3.6, 1.4, 0.2], [7.0, 3.2, 4.7, 1.4], [6.4, 3.2, 4.5, 1.5], [6.9, 3.1, 4.9, 1.5], [6.3, 3.3, 6.0, 2.5], [5.8, 2.7, 5.1, 1.9] ]`

`y = [ 0, 0, 0, 0, 0, 1, 1, 1, 2, 2 ]`

`X_test (Test Data) = [ [5.2, 3.4, 1.5, 0.2], [6.5, 3.0, 4.6, 1.5], [6.2, 3.0, 5.2, 2.0] ]`

- Q3)** Viva **[05 marks]**

- Q1)** Write a program to implement DFS traversal for a graph. **[10 marks]**



- Q2)** Write a program to implement a Non-Linear Support Vector Machine (SVM) using the Radial Basis Function (RBF) Kernel. **[20 marks]**

Input Data :

$X = [[1, 2], [2, 3], [5, 5]]$

Class Labels :

$y = [0, 0, 1]$

Test Input : $[4, 4]$

OR

- Q2)** Write a program to implement a Basic Artificial Neural Network (ANN) for predicting outputs from input data as given below. Use different values for the hyperparameters for the learning rate and number of epochs. Find the ideal value for learning rate and # of epochs. **[20 marks]**

Learning rate: 0.1, and number of epoch = 20 (initially).

Input_vectors = $[[3, 1.5], [2, 1], [4, 1.5], [3, 4], [3.5, 0.5], [2, 0.5], [5.5, 1], [1, 1]]$

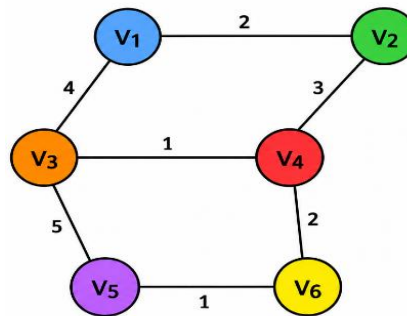
and targets = $[0, 1, 0, 1, 0, 1, 1, 0]$

- Q3)** Viva **[05 marks]**

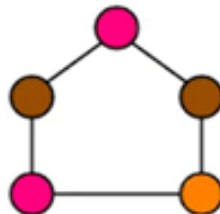
- Q1)** Write a program to implement the A* Search Algorithm to find the shortest path between a source node and a destination node. **[10 marks]**

Source = V_1

Destination = V_6



- Q2)** Write a program to solve the Map Coloring Problem using the Constraint Satisfaction Problem (CSP) approach. **[20 marks]**



OR

- Q2)** Write a program to develop a Tic-Tac-Toe Game in which the computer uses the Minimax Algorithm to play optimally. **[20 marks]**

- Q3)** Viva **[05 marks]**

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination
Lab Course CS-321-VSC-P
Foundation of Artificial Intelligence and Machine Learning

Duration: 3 Hours

Maximum Marks:35

Q1) Write a program to solve the 4-Puzzle Problem using the A* Search Algorithm. **[10 marks]**

Q2) Write a program or report that compares Minimax and Alpha-Beta Pruning on the same game tree. **[20 marks]**

OR

Q2) Write a program to design a Rule-Based Expert System for simple decision-making. **[20 marks]**

Q3) Viva **[05 marks]**

Savitribai Phule Pune University

T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination

Lab Course CS-321-VSC-P

Foundation of Artificial Intelligence and Machine Learning

Duration: 3 Hours

Maximum Marks:35

Q1) Write a program to demonstrate the Hill Climbing Algorithm for finding the maximum value of a simple objective function. **[10 marks]**

Q2) Write a program to implement the Forward Chaining inference mechanism. **[20 marks]**

OR

Q2) Write a program to implement the Decision Tree Classifier for a classification problem. Use the dataset provided with columns: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age, Outcome(is the class label / target variable: 0 or 1). **[20 marks]**

Pregnancies	Glucose	BP	SkinThick	Insulin	BMI	DiabetesPF	Age	Outcome
6	148	72	35	0	33.6	0.627	50	1
1	85	66	29	0	26.6	0.351	31	0
8	183	64	0	0	23.3	0.672	32	1
1	89	66	23	94	28.1	0.167	21	0
0	137	40	35	168	43.1	2.288	33	1
5	116	74	0	0	25.6	0.201	30	0
3	78	50	32	88	31	0.248	26	1
10	115	0	0	0	35.3	0.134	29	0
2	197	70	45	543	30.5	0.158	53	1
8	125	96	0	0	0	0.232	54	1
4	110	92	0	0	37.6	0.191	30	0

Split the dataset into 70:30 – train:test.

Evaluate the model using `accuracy_score()`. Optimize the model's performance by tuning the criterion (default is GINI, use entropy) and `max_depth` (=2, 3 or 4) parameter of the DT Classifier

Q3) Viva **[05 marks]**

Savitribai Phule Pune University

T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination

Lab Course CS-321-VSC-P

Foundation of Artificial Intelligence and Machine Learning

Duration: 3 Hours

Maximum Marks:35

Q1) Write a program to implement Means-End Analysis for solving a goal-based problem **[10 marks]**

Q2) Write a program to implement the Random Forest Classifier for classification tasks. **[20 marks]**

```
dataset = { 'Study_Hours': [2, 3, 4, 5, 6, 7, 1, 2, 5, 8],
            'Attendance': [60, 65, 70, 75, 80, 85, 55, 62, 78, 90],
            'Assignment_Score': [55, 60, 65, 70, 75, 80, 50, 58, 72, 85],
            'Result': ['Fail', 'Fail', 'Pass', 'Pass', 'Pass',
                       'Pass', 'Fail', 'Fail', 'Pass', 'Pass'] }
```

OR

Q2) Write a program to demonstrate hyperplane classification using a Support Vector Machine on the iris dataset. Create a Pipeline containing a StandardScaler and a LinearSVC (with c =1 and use hinge loss). **[20 marks]**

No.	Sepal Length	Sepal Width	Petal Length	Petal Width
1	5.1	3.5	1.4	0.2
2	4.9	3.0	1.4	0.2
3	4.7	3.2	1.3	0.2
4	4.6	3.1	1.5	0.2
5	5.0	3.6	1.4	0.2
6	7.0	3.2	4.7	1.4
7	6.4	3.2	4.5	1.5
8	6.9	3.1	4.9	1.5
9	5.5	2.3	4.0	1.3
10	6.5	2.8	4.6	1.5

Q3) Viva **[05 marks]**

Savitribai Phule Pune University

T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination

Lab Course CS-321-VSC-P

Foundation of Artificial Intelligence and Machine Learning

Duration: 3 Hours

Maximum Marks:35

Q1) Write a program or report that compares Minimax and Alpha-Beta Pruning on the same game tree. **[10 marks]**

Q2) Write a program to implement the Backward Chaining inference mechanism. **[20 marks]**

OR

Q2) Write a program to implement the Random Forest Classifier for classification tasks. Use the dataset provided: car_evaluation.csv (7 columns/features and 10 rows). All the features are of categorical data type, given by buying, maint, doors, persons, lug_boot, safety and class. Class is the target variable. Rename column names as ['buying', 'maint', 'doors', 'persons', 'lug_boot', 'safety', 'class']. Check the frequency counts of categorical variables. Check for missing values. Drop 'class' feature to create X and let y = 'class' variable. Split the dataset as 67:33% - train:test. Encode all the categorical variables. Implement the RF classifier with default parameters. Display the accuracy of the RF classifier. **[20 marks]**

No.	buying	maint	doors	persons	lug_boot	safety	class
1	vhigh	vhigh	2	2	small	low	unacc
2	vhigh	vhigh	2	2	small	med	unacc
3	vhigh	vhigh	2	2	small	high	unacc
4	vhigh	vhigh	2	2	med	low	unacc
5	vhigh	vhigh	2	2	med	med	unacc
6	vhigh	vhigh	2	2	med	high	unacc
7	vhigh	vhigh	2	2	big	low	unacc
8	vhigh	vhigh	2	2	big	med	unacc
9	vhigh	vhigh	2	2	big	high	unacc
10	vhigh	vhigh	2	4	small	low	unacc

Q3) Viva **[05 marks]**

Savitribai Phule Pune University

T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination

Lab Course CS-321-VSC-P

Foundation of Artificial Intelligence and Machine Learning

Duration: 3 Hours

Maximum Marks:35

Q1) Write a program to implement Breadth First Search (BFS) for the Monkey Banana Problem. **[10 marks]**

Q2) Write a program or report to compare Linear SVM and Non-Linear SVM (RBF Kernel). **[20 marks]**

$X = [[1, 2], [2, 3], [3, 1], [4, 2], [5, 5], [6, 6], [7, 4], [8, 5]]$

$Y = [0, 0, 0, 0, 1, 1, 1, 1]$

OR

Q2) Write a program to implement an Artificial Neural Network (ANN) for a Linear Regression problem. **[20 marks]**

Q3) Viva **[05 marks]**

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination
Lab Course CS-321-VSC-P
Foundation of Artificial Intelligence and Machine Learning

Duration: 3 Hours

Maximum Marks:35

Q1) Write a program to solve the Water Jug Problem using BFS. **[10 marks]**

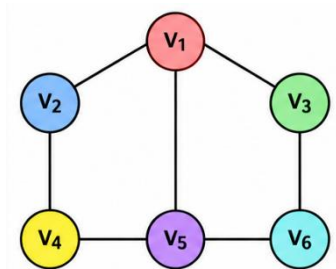
Q2) Write a program to implement a Voting Classifier by combining multiple machine learning models such as Logistic Regression and Decision Tree. **[20 marks]**

OR

Q2) Write a program to develop a simple Knowledge Graph representing relationships among entities. **[20 marks]**

Q3) Viva **[05 marks]**

- Q1)** Write a program to implement BFS traversal for a graph. **[10 marks]**



- Q2)** Write a program or report to compare Supervised Learning and Unsupervised Learning techniques. **[20 marks]**

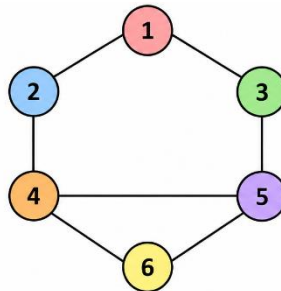
OR

- Q2)** Write a program to demonstrate basic Ontological Engineering concepts using classes, subclasses and relationships. **[20 marks]**

- Q3)** Viva **[05 marks]**

Q1) Write a program to solve a state-space search problem using DFS. **[10 marks]**

Start vertex - 1



Q2) Write Python program to demonstrate the following objective functions used in clustering algorithms. (Any 2 of the following) **[20 marks]**

1. K-Means Inertia (Within-Cluster Sum of Squares – WCSS)
2. Gaussian Mixture Model (GMM) Log-Likelihood
3. Silhouette Coefficient
4. Davies-Bouldin Index

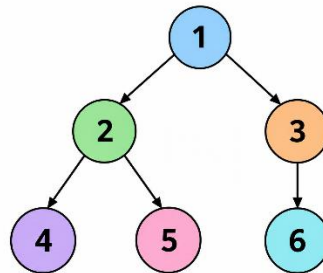
Using matplotlib to plot the Elbow curve, Silhouette scores, and Davies-Bouldin scores across different values of K (on x-axis).

OR

Q2) Write a program to build a Knowledge Base using logical rules and facts. **[20 marks]**

Q3) Viva **[05 marks]**

- Q1) Write a program to implement Depth Limited Search (DLS). [10 marks]



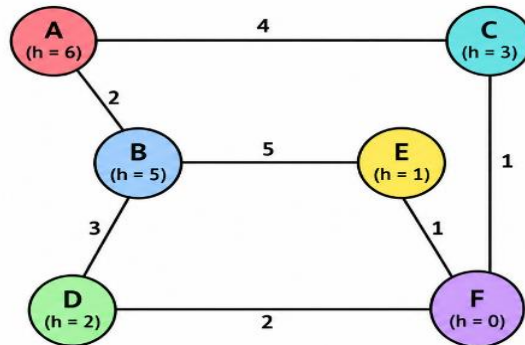
- Q2) Write a program to implement the Backward Chaining inference mechanism. [20 marks]

OR

- Q2) Write a program to implement the K-Means Clustering Algorithm for grouping data into clusters. Use the data set provided in country_data.csv. The columns are: country nm, child_mortality, exports, health, imports, income, inflation, life_expectancy, total_fer, gdp. Normalize the data set for KMeans clustering (hint: use StandardScaler()). Apply KMeans with number of clusters = 3. Use different values for k from 1 to 10, use Elbow method to find the ideal value of 'k'.

- Q3) Viva [05 marks]

- Q1)** Write a program to implement the Best First Search Algorithm using heuristic values to find a path from a start node to a goal node. **[10 marks]**



- Q2)** Write a program to implement Propositional Logic and evaluate logical expressions using operators such as AND, OR, and IMPLIES. **[20 marks]**

OR

- Q2)** Write a program to calculate and demonstrate the Euclidean Distance between two data points in Machine Learning. The program should accept two input vectors from the user and display the calculated Euclidean Distance. **[20 marks]**

Input :

Data Point: 2 3 4

Data Point: 5 7 6

Output :

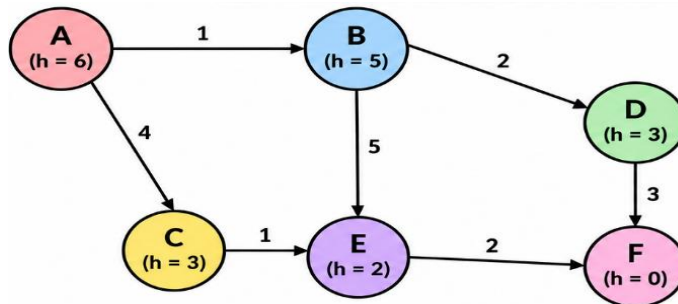
Data Point : [2, 3, 4]

Data Point : [5, 7, 6]

Euclidean Distance = 5.099

- Q3)** Viva **[05 marks]**

- Q1) Write a program to implement the A* Search Algorithm to find the shortest path between a source node and a destination node. [10 marks]



- Q2) Write a program to implement an Artificial Neural Network (ANN) for a Linear Regression problem. [20 marks]

OR

- Q2) Write a program to visualize the decision boundaries created by an SVM classifier. [20 marks]

- Q3) Viva [05 marks]

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination
Lab Course CS-321-VSC-P
Foundation of Artificial Intelligence and Machine Learning

Duration: 3 Hours

Maximum Marks:35

Q1) Write a program to solve the 8-Puzzle Problem using the A* Search Algorithm. **[10 marks]**

Q2) Write a program to build a Convolutional Neural Network (CNN) for image recognition.(Orange vs Apple) **[20 marks]**

OR

Q2) Write a program to analyze the performance of a Support Vector Machine (SVM) on a classification dataset. **[20 marks]**

Q3) Viva **[05 marks]**

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination
Lab Course CS-321-VSC-P
Foundation of Artificial Intelligence and Machine Learning

Duration: 3 Hours

Maximum Marks:35

Q1) Write a program to demonstrate the Hill Climbing Algorithm for finding the maximum value of a simple objective function. **[10 marks]**

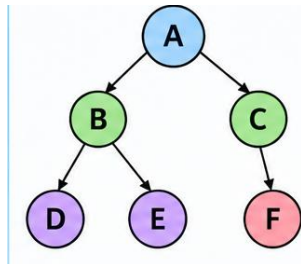
Q2) Write a program to implement a Convolutional Neural Network (CNN) for basic object detection.(person, car, or dog) **[20 marks]**

OR

Q2) Write a program or report to compare Supervised Learning and Unsupervised Learning techniques. **[20 marks]**

Q3) Viva **[05 marks]**

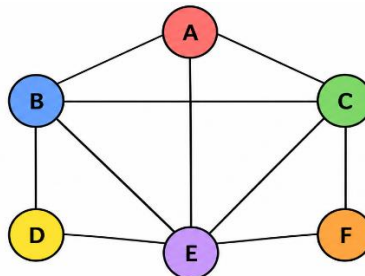
- Q1) Write a program implement Iterative Deepening Search (IDS). [10 marks]



- Q2) Write a program to implement the Gaussian Naïve Bayes Classifier for classifying numerical data. [20 marks]

OR

- Q2) Write a program to solve the Map Colouring Problem using the Constraint Satisfaction Problem (CSP) approach. [20 marks]



- Q3) Viva [05 marks]

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination
Lab Course CS-321-VSC-P
Foundation of Artificial Intelligence and Machine Learning

Duration: 3 Hours

Maximum Marks:35

Q1) Write a program to solve the Tower of Hanoi Problem using state-space representation. **[10 marks]**

Q2) Write a program to solve the N-Queens Problem using the Backtracking Algorithm. **[20 marks]**

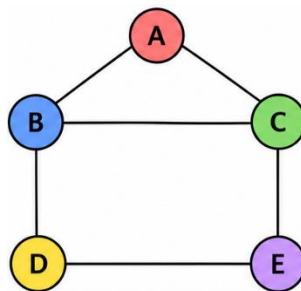
OR

Q2) Write a program to design a Rule-Based Expert System for simple decision-making. **[20 marks]**

Q3) Viva **[05 marks]**

Q1) Write a program to demonstrate the Hill Climbing Algorithm for finding the maximum value of a simple objective function. ($F(x) = -x^2 + 4x$) **[10 marks]**

Q2) Write a program to solve the Map Colouring Problem using the Constraint Satisfaction Problem (CSP) approach. **[20 marks]**

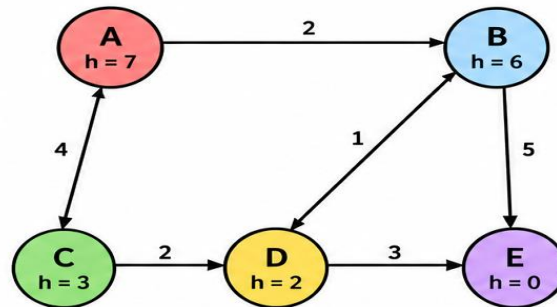


OR

Q2) Write a program or report to compare Supervised Learning and Unsupervised Learning techniques. **[20 marks]**

Q3) Viva **[05 marks]**

- Q1) Write a program to implement the A* Search Algorithm to find the shortest path between a source node and a destination node. [10 marks]



- Q2) Write a program to implement the K-Means Clustering Algorithm for grouping data into clusters. [20 marks]

OR

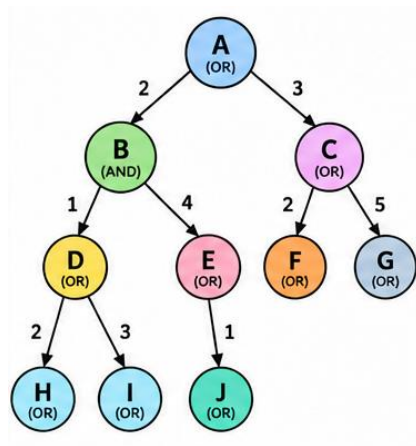
- Q2) Write a program to classify images using a Convolutional Neural Network (CNN) on the MNIST data. Normalize the input data and reshape for CNN. Split the dataset as 80:20 – train:test. Define the CNN model with 2 convolution layers, dropout of 20%, max pool size = 2X2, padding = 1, learning rate = 0.01, epochs= 10, Adam optimizer and cross entropy loss. Evaluate the model on Test Set and print its accuracy. [20 marks]

No.	Image Size	Label (Digit)	Pixel Value Range	Description
1	28 × 28	0	0–255	Handwritten digit 0
2	28 × 28	1	0–255	Handwritten digit 1
3	28 × 28	2	0–255	Handwritten digit 2
4	28 × 28	3	0–255	Handwritten digit 3
5	28 × 28	4	0–255	Handwritten digit 4
6	28 × 28	5	0–255	Handwritten digit 5
7	28 × 28	6	0–255	Handwritten digit 6
8	28 × 28	7	0–255	Handwritten digit 7
9	28 × 28	8	0–255	Handwritten digit 8
10	28 × 28	9	0–255	Handwritten digit 9

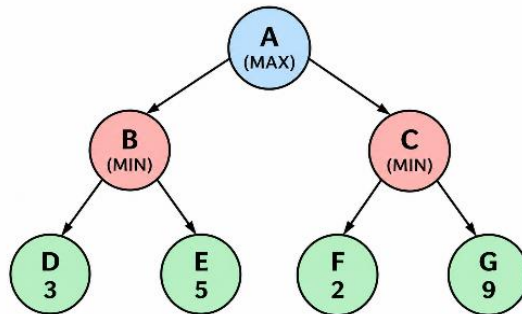
- Q3) Viva [05 marks]

Q1) Write the program to implement the AO* Algorithm

[10 marks]



Q2) Write a program to implement the Minimax Algorithm for a two-player game. [20 marks]



OR

Q2) Write a program to implement the Forward Chaining inference mechanism. [20 marks]

Q3) Viva

[05 marks]